Assignment 3

Due date: TBA.
Total: 30 points
Coverage: Chapter 4: Generating Random Variables and Chapter 5: Monte Carlo Simulation and Variance Reduction.

Question	Format	Points
1	8 multiple-choice questions on random variable generation	8
2	8 multiple-choice questions on simulation and variance reduction	8
3	6 true/false questions	6
4	2 short-answer parts with four prompts each	8
Total		**30**

Assignment PDF | Response template (.qmd)

Submission

Use **R Markdown** or **Quarto**. Submit your **source file** (`.Rmd` or `.qmd`) and **rendered PDF** to the **Assignment 3** folder on **iCollege**. Name both files `A3_LastName_FirstName` with the appropriate extension, and include your name in the PDF.
Use the response template, or follow its answer table and question labels.

- **Questions 1-2:** Select one best answer, A, B, C, or D.
- **Question 3:** Write **True** or **False**.
- **Question 4:** Give a short explanation or calculation for each prompt.

Each item is worth **1 point**. Only Question 4 requires explanations or calculations.
R code and plots are optional. If you use R, include the code in your source file. Before generating random values, set a seed (e.g., `set.seed(8670)`) and keep that line in your code. Check that your source renders to PDF without errors.

Note: The questions cover the main material in Chapters 4 and 5. Supplementary sections, proofs, and derivations of optimal variance-reduction coefficients are not required.

Question 1: Generating Random Variables [8]

Select the **best answer**. Each item is worth **1 point**.

1a)

Consider the following R code:

```r
x <- numeric(100)
for (i in seq_len(100)) {
  set.seed(8670)
  x[i] <- runif(1)
}
```

What is the main problem with using `x` as 100 independent uniform draws?
A. The values must all be zero.
B. The values are sampled without replacement.
C. Resetting the seed repeats the same first draw in every iteration.
D. A uniform generator cannot be used inside a loop.

1b)

Let $U \sim \text{Uniform}(0, 1)$. Which expression generates an exponential random variable with **mean 5**?
A. $-\log(1-U)/5$
B. $-5\log(1-U)$
C. $\exp(-5U)$
D. $5U$

1c)

The Beta$(2, 2)$ density is $f(x) = 6x(1-x)$ for $0 < x < 1$, and zero otherwise. Its height at $x = 0.5$ is 1.5. Which statement is correct?
A. It cannot be a density because its height exceeds 1.
B. $P(X = 0.5) = 1.5$.
C. The area under the curve is 1.5.
D. It is a valid density: its total area is 1, even though its height can exceed 1.

1d)

An unnormalized target is $h(x) = x(1-x)$ on $(0, 1)$, with total area $1/6$. Which formula turns it into a normalized density?
A. $f(x) = 6h(x)$
B. $f(x) = h(x)/6$
C. $f(x) = h(x) - 1/6$
D. $f(x) = h(x)$

1e)

For the target in 1d), compare two rejection samplers with a uniform proposal on $(0, 1)$:

- normalized scale: target $f(y) = 6y(1 - y)$, envelope height 1.5;
- unnormalized scale: target $h(y) = y(1 - y)$, envelope height 0.25.

Suppose both samplers use the same proposal Y and the same draw $U \sim \text{Uniform}(0, 1)$, independent of Y. How do their acceptance decisions compare?
A. The normalized version accepts six times as often.
B. The unnormalized version cannot be used for rejection sampling.
C. Both accept exactly when $U \leq 4Y(1 - Y)$.
D. Both accept exactly when $U \leq Y(1 - Y)$.

1f)

Use the normalized sampler in 1e). For one proposal, $Y = 0.75$ and $U = 0.60$. Its acceptance ratio is $f(Y)/1.5 = 0.75$. What should the sampler do?
A. Accept and store 0.75.
B. Accept and store 0.60.
C. Reject because $U < 0.75$.
D. Accept and store the density height 1.125.

1g)

For the sampler in 1e), a proposal is plotted at (Y, V) with $V = 1.5U$, where U is uniform on $(0, 1)$ and independent of Y. What does the point's height V represent?
A. The probability that the target equals Y.
B. The value of the target density $f(Y)$.
C. A second accepted draw from the target distribution.
D. An auxiliary random height between zero and the envelope, used to decide whether to accept Y.

1h)

An envelope lies above the target at 100 grid points. What else must be checked before using it for rejection sampling? Assume the proposal covers the target support.
A. The number of proposals is at least 100.
B. The envelope is at least as high as the target throughout its support.
C. The envelope has total area exactly 1.
D. Every proposal is accepted.

Question 2: Simulation and Variance Reduction [8]

Select the **best answer**. Each item is worth **1 point**.

2a)

Why use simulation to estimate an event probability when a direct calculation is difficult?
A. It guarantees that the chosen model matches the real world.
B. It removes all uncertainty after a single run.
C. It generates repeated outcomes from a model and estimates how often the event occurs.
D. It replaces the event probability by the mean of the model's inputs.

2b)

A shop has 12 muffins each day. The vector `demand` contains the number of customers wanting one muffin on each simulated day. Which R expression estimates the probability that **at least one customer cannot be served**?
A. `mean(demand > 12)`
B. `mean(demand)`
C. `sum(demand > 12)`
D. `mean(demand >= 12)`

2c)

Two simulation runs use the same model and number of days, but independent random draws. Why can their estimated shortage probabilities differ?
A. The stock level must have changed.
B. The model's true shortage probability changes after each run.
C. One of the runs must contain a coding error.
D. Different finite samples contain different numbers of shortage days.

2d)

A simulation averages independent outputs $Y_1, \ldots, Y_N$. Which statement correctly distinguishes the output standard deviation (SD) from the Monte Carlo standard error (MCSE)?
A. Both measure only the spread of individual Y_i values.
B. SD describes the spread of individual outputs; MCSE describes the uncertainty of their average due to simulation.
C. SD describes the uncertainty of the average; MCSE describes individual outputs.
D. MCSE measures how wrong the model's assumptions are.

2e)

With 1,000 independent simulated days, an estimated probability has MCSE about 0.02. With the same model and method, what MCSE should we expect using 4,000 days?
A. About 0.01.
B. About 0.005.
C. About 0.02.
D. Exactly zero.

2f)

What is the purpose of variance reduction?
A. Change the quantity of interest to one with a smaller numerical value.
B. Make real customer demand less variable by simulating more days.
C. Estimate the same quantity with less variation across repeated runs for a similar amount of computation.
D. Select the random seed that happens to give the closest answer.

2g)

An antithetic method generates 500 independent $U_j \sim \text{Uniform}(0,1)$ draws and forms the 500 pair averages $A_j = \{e^{U_j} + e^{1-U_j}\}/2$. Which values are the independent units used in $\text{sd}(\cdot)/\sqrt{N}$ to estimate MCSE?
A. The 1,000 separate function values, treating all of them as independent.
B. The 500 pair averages, so MCSE is $\text{sd}(A)/\sqrt{500}$.
C. Only the first pair average.
D. The two values in any single pair.

2h)

For $U \sim \text{Uniform}(0,1)$, a control-variate method uses $Y_{\text{adj}} = e^U - c(U - 0.5)$, with a fixed coefficient c. Why does this adjustment preserve the target mean $\mathbb{E}[e^U]$?
A. The adjusted output equals e^U for every draw.
B. Any constant can replace 0.5 without changing the mean.
C. The adjustment always makes the output positive.
D. The subtracted term has mean zero because $\mathbb{E}[U] = 0.5$.

Question 3: Interpreting Simulation [6]

Write **True** or **False** for each statement. Each item is worth **1 point**. No explanation is required.

3a)

Setting the seed once at the start allows the same simulation to be reproduced when the code and computing environment are unchanged.

3b)

Increasing the number of simulated days makes real customer demand less variable from day to day.

3c)

In rejection sampling, increasing a valid envelope by a constant factor greater than 1 preserves the distribution of accepted values but lowers the acceptance rate.

3d)

Pairing U and $1 - U$ must reduce Monte Carlo variance for every possible function of U.

3e)

A small MCSE guarantees that the model assumptions accurately describe the real world.

3f)

A comparison of simulation methods should examine the variability of repeated estimates of the same quantity, using a similar amount of computation per run.

Question 4: Simulating Daily Demand [8]

Each item is worth **1 point**. Label your answers **4a.1-4a.4** and **4b.1-4b.4**. Keep explanations to one sentence and show the requested calculations.

4a) Estimating the shortage probability [4]

A shop prepares 12 muffins each morning and has 20 potential customers per day. Assume that each customer independently wants one muffin with probability 0.6 and that demand is independent across days.

Let D be the number of customers wanting a muffin in one day. We want to estimate $p = P(D > 12)$, the probability that at least one customer cannot be served.

1. The mean demand is $20 \times 0.6 = 12$. Why might 12 muffins still not be enough?
2. Describe how to simulate one day's demand D using independent uniform draws. Specify the number of draws and how each draw determines whether a customer wants a muffin. A line of R code is also acceptable.
3. Complete the indicator rule: $Y = 1$ if _____, and $Y = 0$ otherwise. Write the condition as an inequality involving D.
4. A run of 1,000 independent days includes 430 shortage days. Compute $\hat{p}$ as a fraction and a decimal.

4b) Variation between runs [4]

A second run with the same assumptions and 1,000 new simulated days produces 397 shortage days.

1. The two estimates are 0.430 and 0.397. Why can they differ even if both programs are correct?
2. For the first run, use $\hat{p} = 0.430$ and $N = 1{,}000$ to compute $\widehat{\text{MCSE}} = \sqrt{\hat{p}(1-\hat{p})/N}$. Give the value to **four decimal places**.
3. If N increases to 4,000 and $\hat{p}$ stays close to 0.430, what approximate MCSE do you expect? Give the value to **four decimal places**.
4. When only the number of simulated days increases, complete each statement with **decreases**, **stays the same**, or **increases**: the spread of $\hat{p}$ across runs _____; the day-to-day spread of demand D under the model _____.